Polar Coordinates & Polar Graphs

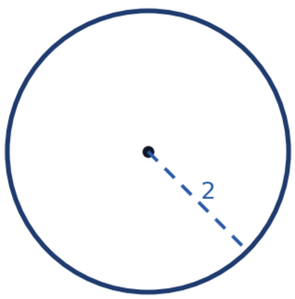
Converting between polar and rectangular coordinates

1 Convert the polar point $\left(4, \frac{2\pi}{3}\right)$ to rectangular coordinates.

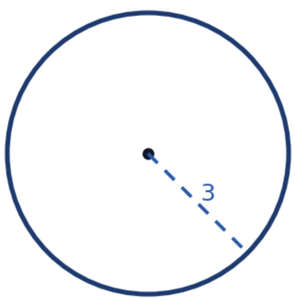
2 Convert the rectangular point $(-3, 3)$ to polar coordinates with $r > 0$ and $0 \le \theta < 2\pi$.

Converting polar equations to rectangular form

3 Convert the polar equation $r = 4\cos\theta$ to rectangular form, and identify the curve.

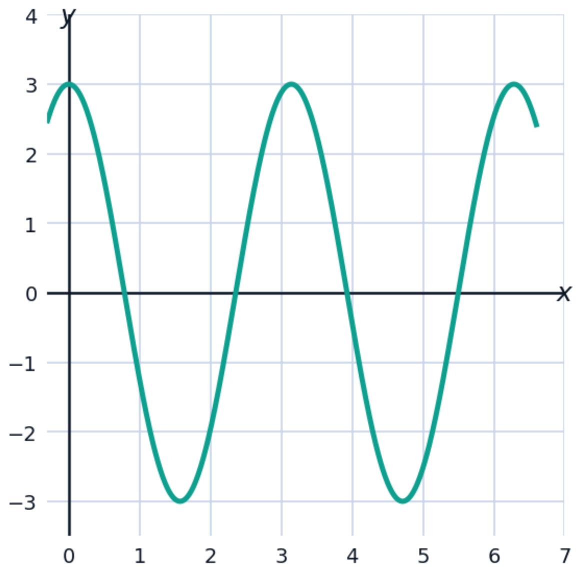


4 Convert the rectangular equation $x^2 + y^2 = 9$ to polar form.



Visualizing a rectangular (r vs. θ) graph of a polar rose

5 The graph shows $r = 3\cos(2\theta)$ plotted in rectangular form, with θ on the horizontal axis (in radians) and r on the vertical axis — a standard technique for analyzing a polar rose curve before sketching it.

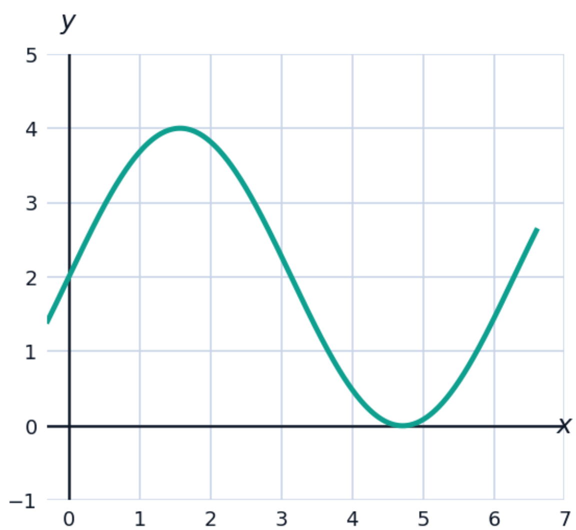


a State how many petals the actual polar rose $r = 3\cos(2\theta)$ has, using the rule for $r = a\cos(n\theta)$ with even n.

b Read off the graph: at which values of θ in $[0, 2\pi)$ does $r = 0$ (these are the angles where the rose curve passes through the pole)?

Symmetry and special polar curves

6 Identify the type of curve represented by $r = 2 + 2\sin\theta$ and state its key feature.



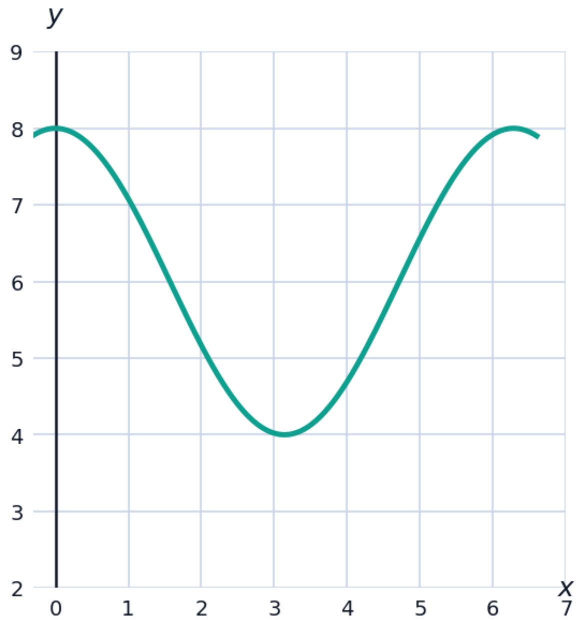
7 Determine whether the graph of $r = 3\cos\theta$ is symmetric about the polar axis (the x-axis), and justify your answer.

Multiple representations of the same polar point

8 Give two other polar representations of the point $\left(5, \frac{\pi}{4}\right)$: one with a negative r-value, and one with a different positive angle (adding 2π).

Applications of polar curves

9 A radar station models the range of its signal by the polar curve $r = 6 + 2\cos\theta$ (in km), where θ is measured from due east. Find the maximum and minimum detection range, and the direction of each.

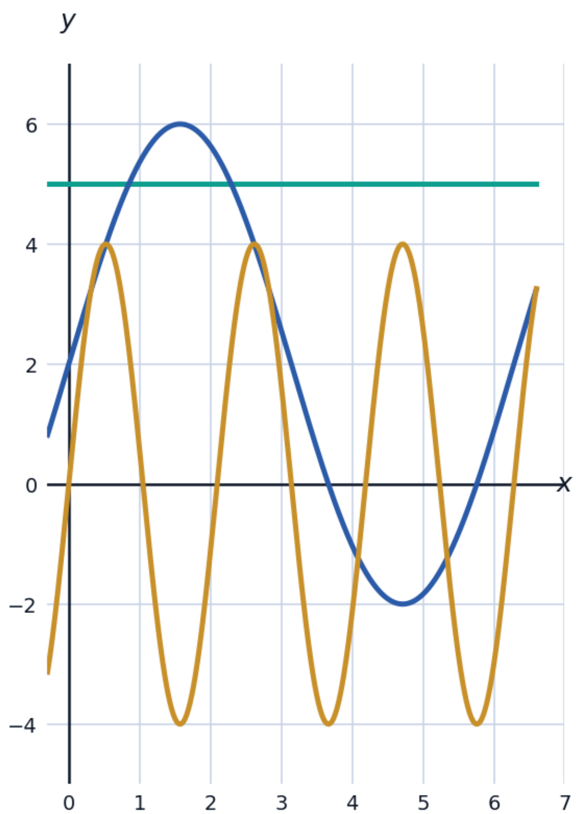


Distance between points in polar form

10 Find the distance between the polar points $\left(3, \frac{\pi}{6}\right)$ and $\left(5, \frac{2\pi}{3}\right)$ by first converting both to rectangular coordinates.

Identifying polar curve families

11 Classify each polar equation as a circle, cardioid, limaçon (with inner loop), or rose, and briefly justify: (a) $r = 5$, (b) $r = 2 + 4\sin\theta$, (c) $r = 4\sin(3\theta)$.



Points of intersection of polar curves

12 Find the values of θ in $[0, 2\pi)$ where the polar curves $r = 2$ and $r = 4\cos\theta$ intersect.

Area in polar coordinates (introductory)

13 Explain, without evaluating an integral, why the polar area formula $A = \frac{1}{2}\int_a^b r^2\,d\theta$ uses r^2 rather than r, by comparing it to the area of a circular sector $A = \frac{1}{2}r^2\theta$.

14 A circular sector has radius 6 and central angle $\frac{\pi}{3}$. Use the sector-area formula $A = \frac{1}{2}r^2\theta$ to find its area.